PRODUCT DESIGN · REQUIREMENTS

v1.0 · 28 October 2026

Coffee Shop *Website*

UI/UX Design Requirements

A Liquid-Glass-inspired design system and screen specification for a world-class café web experience: translucent, tactile, warm and alive.



Liquid Glass UI

Translucent layers, specular rims, floating bars that shrink on scroll.

Brewed palette

Espresso, crema, oat, caramel, matcha & berry cold-brew tokens, light + dark.

Alive by design

Steam heroes, pour-fill cups, latte-art loaders, brew-progress tracking.

Crema Menu Rewards Stores Our story Order ahead

↑ FLOATING GLASS NAVIGATION, AS SPECIFIED IN CMP-NAV

DOCUMENT	VERSION · DATE	OWNER	AUDIENCE
UI/UX Requirements	v1.0 · 28 Oct 2026	Tanvir Hasan	Frontend & Design team

00 · ABOUT THIS DOCUMENT

Read me first

This specification tells design and frontend exactly **what** to build, **how it should feel**, and **how we will judge it done**. Every requirement carries an ID and a priority so it can be traced from Figma to pull request to QA.

Document control

FIELD	VALUE
Title	Coffee Shop Website: UI/UX Design Requirements
Version	1.0 (baseline for design kickoff)
Date	28 October 2026
Owner	Tanvir Hasan
Audience	Frontend developers, UI/UX & motion designers
Status	Approved for design exploration
Working brand	“Crema” (placeholder name, swap freely)
Design language	Inspired by Apple iOS 26 “Liquid Glass”, adapted for the web

Priority scale (MoSCoW)

LABEL	MEANING
MUST	Required for v1 launch. No release without it.
SHOULD	Strongly expected in v1; drop only with owner sign-off.
COULD	Delight layer / v1.1 candidate; build behind a feature flag.

Contents

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Requirement ID scheme

PREFIX	AREA
BR-	Brand & principles
TK-	Design tokens
LG-	Liquid Glass implementation
CMP-	Components
SCR-xx-	Screens (HOME, MENU, CUST, CART, TRK, LOY, LOC, ABT, PRE)
FX-	Creative coffee features
ANM- / MI-	Animation / micro-interactions
RSP- / A11Y- / PERF-	Responsive, accessibility, performance

Warm glass, slow coffee

The site should feel like holding a glass of iced latte up to morning sunlight: layered, translucent, warm, a little playful. Liquid Glass gives us depth and light; coffee gives us colour, texture and ritual.



Warm · Translucent · Tactile · Calm · Playful, not childish · Crafted · Sensory · Premium but friendly

Design principles

ID	PRINCIPLE	PRIORITY
BR-01	**Content first, glass second.** Glass is a *layer that floats above content* (nav, tab bar, sheets, controls), never the content itself. Product photography must stay vivid and unblurred.	MUST
BR-02	**Depth with purpose.** Max 3 visible depth layers at once: content → floating glass controls → transient overlays (sheets/toasts). Each layer gets one elevation + blur token.	MUST
BR-03	**Coffee is the colour system.** All accents map to real ingredients (caramel, matcha, berry, nitro). No arbitrary blues/greys outside the token set.	MUST
BR-04	**Alive, never noisy.** Ambient motion (steam, foam, shimmer) runs at low amplitude and stops when off-screen or with reduced motion.	MUST
BR-05	**Tactile feedback everywhere.** Every press, toggle and add-to-cart has a visible response <100 ms (scale, highlight, haptic on supported devices).	MUST
BR-06	**Adaptive by default.** Light ↔ dark, morning ↔ night themes and system contrast settings change tokens, not components.	SHOULD
BR-07	**Concentric geometry.** Corner radii are concentric: inner radius = outer radius − padding, giving the iOS-style nested "capsule in capsule" look.	SHOULD
BR-08	**One hero moment per screen.** Each screen gets exactly one signature animation (e.g. cup fill, brew ring); everything else is subtle.	SHOULD

> **Tone of voice:** short, warm, sensory microcopy. "Your oat flat white is brewing ☕", not "Order status: processing". Use the customer's first name sparingly (order ready, rewards).

Morning Brew palette

Light theme is the default from 06:00 to 17:59 local time (see FX-05). Ratios below are measured against `--color-bg #FBF6EF`. Swatches marked <4.5:1 are **fill-only** colours and must carry Espresso text.

#2B1A12	#6B4430	#7A6455	#FBF6EF
15.5:1 on bg	7.8:1 on bg	5.2:1 on bg	1.0:1 on bg
Espresso 900 `--color-text` Primary text, icons, dark CTAs	**Roast 600** `--color-text-2` Secondary text, captions	**Bean Muted** `--color-text-muted` Meta, placeholders (≥4.5:1)	**Oat Milk 50** `--color-bg` App background

#FFFFFF	#F4E6D4	#E9C9A0	#D98A3D
1.1:1 on bg	1.1:1 on bg	1.5:1 on bg	2.6:1 on bg
Steamed Milk `--color-surface` Cards, sheets (under glass)	**Crema 100** `--color-surface-2` Grouped sections, inputs	**Crema 300** `--color-crema` Foam layer, highlights, borders	**Caramel 500** `--color-primary` Primary CTA fill (Espresso text)

#A85A12	#8DBF6A	#3F6B26	#9E2F63
4.7:1 on bg	2.0:1 on bg	5.8:1 on bg	6.4:1 on bg
Caramel 700 `--color-primary-ink` Links, active labels, price	**Matcha 500** `--color-matcha` Fresh/vegan/success fill	**Matcha 700** `--color-success` Success text & icons	**Berry Cold Brew 600** `--color-berry` Seasonal accent, badges, Must

#3D3A8C	#B42318
8.9:1 on bg	6.1:1 on bg
Nitro Plum 700 `--color-nitro` Cold drinks, info, links alt	**Cherry 700** `--color-danger` Errors, destructive

TK-C01 · Must: implement every colour as a CSS custom property on `:root` and `[data-theme="dark"]`; components reference semantic tokens only (never raw hex).

TK-C02 · Must: Caramel 500 and Matcha 500 are *fills*. Text on them is Espresso 900 (6.1:1 / 10.0:1). Never use them as text colour on light backgrounds.

02 · COLOUR · DARK THEME

Night Roast palette

Dark theme follows `prefers-color-scheme`, the manual toggle, or Night Decaf mode (18:00–05:59). Accents are lifted to "glow" values so they stay vivid on roast-black. Ratios measured against `#140E0B`.

`#F7EEE3` 16.7:1 on bg **Steam 50** `--color-text` Primary text	`#CDB9A6` 10.1:1 on bg **Latte 300** `--color-text-2` Secondary text	`#9C8676` 5.5:1 on bg **Husk 500** `--color-text-muted` Meta text (≥5:1)	`#140E0B` 1.0:1 on bg **Night Roast 950** `--color-bg` App background
`#211813` 1.1:1 on bg **Roast Surface 900** `--color-surface` Cards & sheets	`#2E221B` 1.2:1 on bg **Roast Surface 800** `--color-surface-2` Inputs, grouped	`#EBC9A0` 12.2:1 on bg **Crema Glow 300** `--color-crema` Foam, highlights	`#F2A65A` 9.5:1 on bg **Caramel Glow 400** `--color-primary` Primary CTA fill (Night text)
`#F6BE84` 11.5:1 on bg **Caramel Glow 300** `--color-primary-ink` Links, price	`#A8D68A` 11.5:1 on bg **Matcha Neon 300** `--color-success` Success	`#EE86B8` 8.0:1 on bg **Berry Glow 300** `--color-berry` Seasonal accent	`#A9A6FF` 8.7:1 on bg **Nitro Glow 300** `--color-nitro` Cold drinks, info
`#FF8A80` 8.4:1 on bg **Cherry 300** `--color-danger` Errors	`#3A2C23` 1.4:1 on bg **Decaf Ink 700** `--color-border` Hairlines, dividers		

TK-C03 · Must: dark backgrounds are warm (brown-black), never pure `#000`; this keeps glass tints warm and avoids OLED smearing on scroll.

TK-C04 · Should: honour `prefers-contrast: more` by raising glass opacity to 0.85 and switching hairlines to Latte 300.

Gradients, tints & mesh

Gradients are used for **ambient backgrounds and hero surfaces only**, never behind body text without a glass/scrim layer. Mesh blobs are built from 3–4 radial gradients (cheap to render, resolution-independent).



`--grad-morning-pour`
Home hero (day), CTA glow



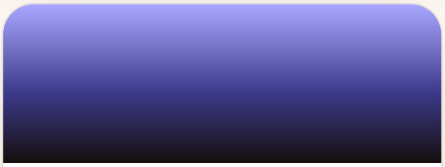
`--grad-night-brew`
Hero (night), footer



`--grad-matcha-latte`
Plant-based category, success



`--grad-berry-cold-brew`
Seasonal specials banner

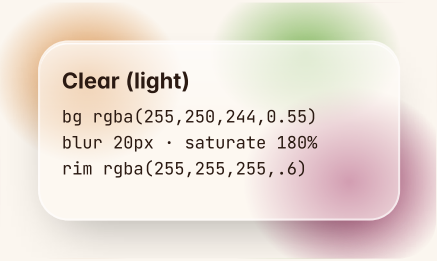


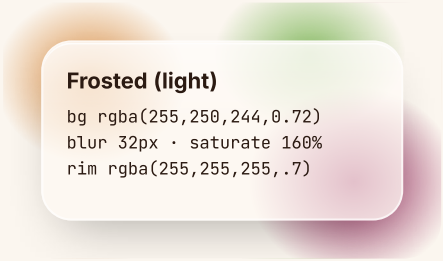
`--grad-nitro-cascade`
Cold-brew category, tracking

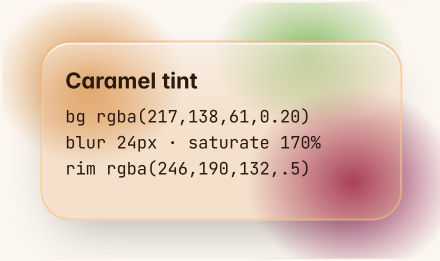


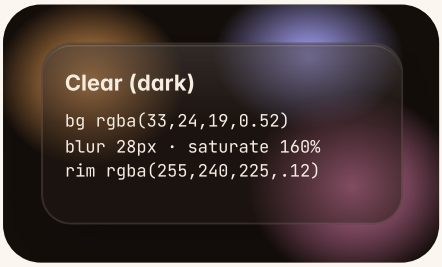
`--grad-caramel-drizzle`
Rewards card, primary hover

Glass tints (rendered over mesh)













Time-of-day themes (FX-05)



Morning Brew · 06:00–11:59
Light theme, Caramel accents, energetic copy, espresso-forward menu order.



Afternoon Chill · 12:00–17:59
Light theme, Matcha & Nitro accents, cold drinks promoted first.



Night Decaf · 18:00–05:59
Dark theme, Berry glow, decaf & desserts first, dimmer motion.

WCAG 2.2 AA contrast matrix

Ratios are computed with the WCAG relative-luminance formula. Targets: body text ≥ 4.5:1, large text (≥ 24 px or 18.66 px bold) ≥ 3:1, UI components & focus indicators ≥ 3:1 (SC 1.4.11).

SAMPLE	FOREGROUND	BACKGROUND	RATIO	RESULT	USAGE
Aa Latte	Espresso 900 text `#2B1A12`	Oat Milk 50 bg `#FBF6EF`	**15.51:1**	AAA ✓	Body text
Aa Latte	Roast 600 text `#6B4430`	Oat Milk 50 bg `#FBF6EF`	**7.83:1**	AAA ✓	Secondary text
Aa Latte	Bean Muted text `#7A6455`	Steamed Milk `#FFFFFF`	**5.55:1**	AA ✓	Meta / captions
Aa Latte	Caramel 700 text `#A85A12`	Oat Milk 50 bg `#FBF6EF`	**4.72:1**	AA ✓	Links, prices
Aa Latte	Espresso 900 text `#2B1A12`	Caramel 500 fill `#D98A3D`	**6.08:1**	AA ✓	Primary button
Aa Latte	Steamed Milk text `#FFFFFF`	Espresso 900 fill `#2B1A12`	**16.68:1**	AAA ✓	Dark button
Aa Latte	Steamed Milk text `#FFFFFF`	Berry 600 fill `#9E2F63`	**6.91:1**	AA ✓	Badges, seasonal CTA
Aa Latte	Matcha 700 text `#3F6B26`	Steamed Milk `#FFFFFF`	**6.28:1**	AA ✓	Success message
Aa Latte	Cherry 700 text `#B42318`	Steamed Milk `#FFFFFF`	**6.57:1**	AA ✓	Error message
Aa Latte	Espresso 900 text `#2B1A12`	Crema 300 fill `#E9C9A0`	**10.58:1**	AAA ✓	Selected chip
Aa Latte	Steam 50 text `#F7EEE3`	Night Roast 950 `#140E0B`	**16.67:1**	AAA ✓	Dark body text
Aa Latte	Latte 300 text `#CDB9A6`	Roast Surface 900 `#211813`	**9.20:1**	AAA ✓	Dark secondary
Aa Latte	Husk 500 text `#9C8676`	Roast Surface 900 `#211813`	**5.05:1**	AA ✓	Dark meta
Aa Latte	Night Roast text `#140E0B`	Caramel Glow 400 `#F2A65A`	**9.46:1**	AAA ✓	Dark primary button
Aa Latte	Berry Glow 300 `#EE86B8`	Night Roast 950 `#140E0B`	**7.96:1**	AAA ✓	Dark accent text
Aa Latte	Nitro Glow 300 `#A9A6FF`	Roast Surface 900 `#211813`	**7.96:1**	AAA ✓	Dark info/link
Aa Latte	Caramel 500 (non-text) `#D98A3D`	Oat Milk 50 bg `#FBF6EF`	**2.55:1**	UI only × text	Icons/focus ring ≥3:1?

TK-C05 · Must: text on glass is validated against the *worst-case* backdrop (darkest/lightest photo region). If it fails, raise tint opacity or add a 12% scrim behind the text. Automate with axe + a Playwright screenshot check on hero images.

TK-C06 · Must: Caramel 500 (2.55:1) is **not** sufficient for focus rings on oat. Focus ring = 2 px Espresso 900 + 2 px offset Caramel 500 halo (dual ring), ≥ 3:1 in both themes.

Type that tastes good

Three free, variable, self-hostable families (Google Fonts / OFL). Fraunces brings a soft, "roasted" serif personality; Inter is the closest free analogue to SF Pro for crisp UI; JetBrains Mono for order codes and receipts.

Aa

Fraunces

Display & editorial

Variable: wght 300–900, opsz 9–144, SOFT 0–100, WONK. Use SOFT 100 for rounded, latte-like terminals.

Aa

Inter

UI, body, numbers

Variable wght 100–900. Enable `cv11`, `ss01`, `tnum` for prices & timers.

#A7

JetBrains Mono

Codes, receipts, tokens

Order numbers (#A-274), pickup codes, ETA countdowns. Max 2 weights.

Type scale (desktop → mobile, px)

TOKEN	FAMILY / WEIGHT	SIZE / LINE	TRACKING	SAMPLE
`display-xl`	Fraunces 600, SOFT 100	64/68 → 40/44	-0.03em	Good mornings, poured.
`display-l`	Fraunces 600	48/52 → 32/36	-0.025em	Seasonal specials
`title-1`	Fraunces 600	32/38 → 26/32	-0.02em	Build your latte
`title-2`	Inter 700	24/30 → 20/26	-0.015em	Your order
`headline`	Inter 600	18/24	-0.01em	Oat Flat White
`body`	Inter 400	16/24	0	Double ristretto, silky oat milk.
`caption`	Inter 600 caps	11/14	0.08em	NEW · SEASONAL
`price`	Inter 650 tnum	17/22	0	$5.40
`code`	JetBrains Mono 600	14/20	0.02em	PICKUP #A-274

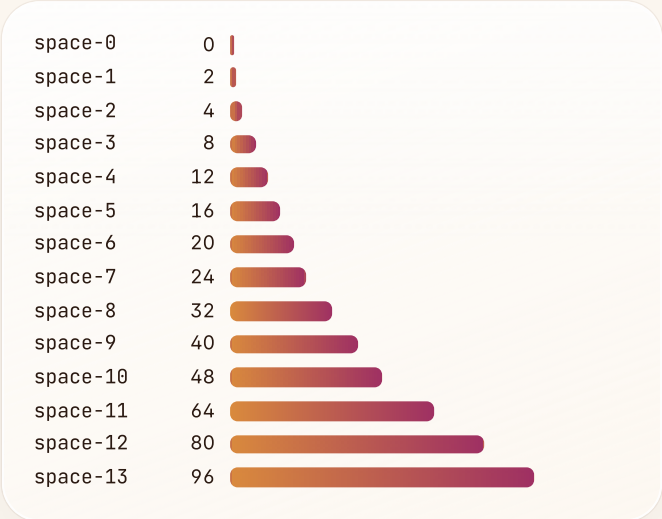
ID	REQUIREMENT	PRIORITY
TK-T01	Self-host fonts as subset WOFF2 (Latin + Latin-Ext), `font-display: swap`, preload only Inter 400–700 variable & Fraunces display. Use `size-adjust` fallback metrics to avoid CLS.	MUST
TK-T02	Fluid type with `clamp()` between 380 px and 1280 px viewports; body never below 16 px on mobile.	MUST

ID	REQUIREMENT	PRIORITY
TK-T03	Line length 45–75 characters for story/about copy; max-width `68ch`.	SHOULD
TK-T04	Respect user font scaling: all sizes in `rem`; layouts must survive 200 % zoom (WCAG 1.4.4) without clipping glass bars.	MUST

The physical system

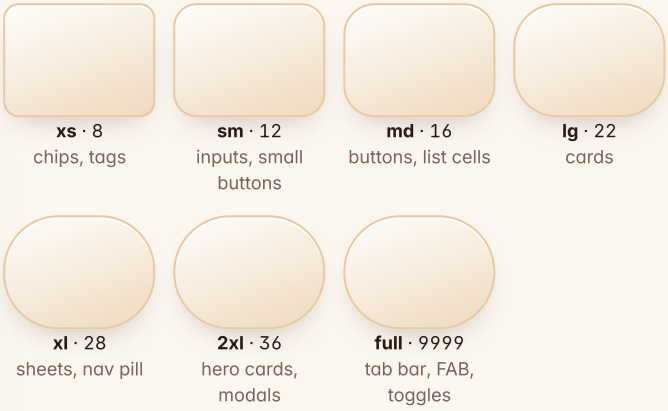
A 4-point grid, concentric radii and a small set of glass depths make every screen feel like it belongs to the same object.

Spacing (px, 4-pt base)



Gutters: 16 (mobile) / 24 (tablet) / 32 (desktop). Section rhythm: 64 → 96 px.

Radius (continuous corners)



TK-R01 · Should: use `corner-shape: squircle` (progressive enhancement, Chromium) with `border-radius` fallback to get Apple-style continuous curves. Nested radius = parent radius − padding.

Elevation

TOKEN	SHADOW	USE
`e0`	`none`	Flat content
`e1`	`0 1px 2px rgb(43 26 18/.06), 0 1px 1px rgb(43 26 18/.04)`	List cells, inputs
`e2`	`0 6px 16px -6px rgb(43 26 18/.18)`	Cards, menu items
`e3`	`0 16px 32px -12px rgb(43 26 18/.28)`	Nav & tab bar (floating)
`e4`	`0 32px 64px -16px rgb(43 26 18/.38)`	Sheets, modals, cup preview

Blur & saturation

TOKEN	BLUR	SATURATE	USE
`glass-thin`	12px	150%	Chips, segmented controls
`glass-regular`	20px	180%	Nav bar, tab bar, cards over photos
`glass-thick`	32px	160%	Sheets, cart drawer
`glass-ultra`	48px	140%	Modal backdrop, scrim

Dark theme: dark shadows use `rgb(0 0 0 / .5)` + 1px top rim `rgb(255 240 225/.10)`.

Tokens as code

Single source of truth: Figma Variables → exported with Tokens Studio / Style Dictionary → `tokens.css` + Tailwind v4 `@theme`. Sample below is normative for names.

```css
/* tokens.css (excerpt) */
:root {
  --color-bg: #FBF6EF;        /* oat milk 50 */
  --color-surface: #FFFFFF;
  --color-surface-2: #F4E6D4;
  --color-text: #2B1A12;      /* espresso */
  --color-text-2: #6B4430;
  --color-primary: #D98A3D;   /* caramel */
  --color-primary-ink: #A85A12;
  --color-matcha: #8DBF6A;
  --color-berry: #9E2F63;
  --color-nitro: #3D3A8C;
  --glass-bg: rgb(255 250 244 / .55);
  --glass-rim: rgb(255 255 255 / .60);
  --glass-hi: rgb(255 255 255 / .80);
  --glass-blur: 20px;
  --glass-sat: 180%;
  --radius-lg: 22px;
  --dur-base: 240ms;
  --ease-standard: cubic-bezier(.2,0,0,1);
}
[data-theme="dark"] {
  --color-bg: #140E0B;
  --color-surface: #211813;
  --color-text: #F7EEE3;
  --color-primary: #F2A65A;
  --glass-bg: rgb(33 24 19 / .52);
  --glass-rim: rgb(255 240 225 / .12);
  --glass-blur: 28px;
}
@media (prefers-reduced-transparency: reduce) {
  :root { --glass-bg: var(--color-surface); --glass-blur: 0px; }
}
```

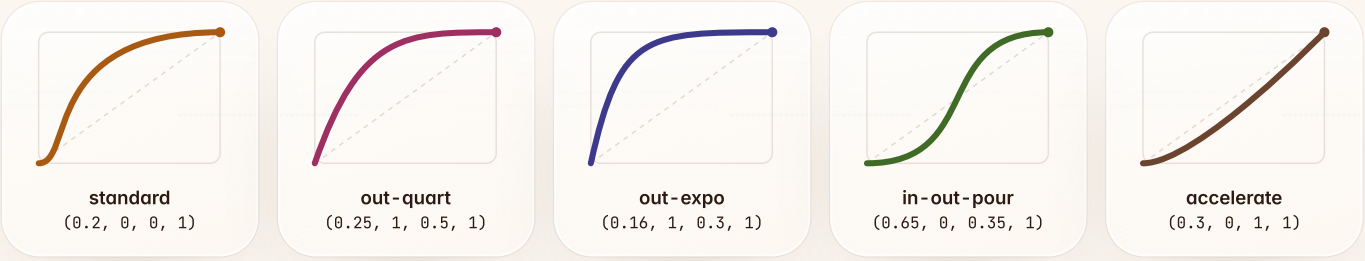
```css
/* app.css (Tailwind v4) */
@import "tailwindcss";
@theme {
  --font-display: "Fraunces", ui-serif, Georgia;
  --font-sans: "Inter", system-ui, -apple-system;
  --font-mono: "JetBrains Mono", ui-monospace;
  --color-espresso: #2B1A12;
  --color-caramel: #D98A3D;
  --color-matcha: #8DBF6A;
  --color-berry: #9E2F63;
  --breakpoint-sm: 380px;
  --breakpoint-md: 768px;
  --breakpoint-lg: 1024px;
  --breakpoint-xl: 1280px;
}
@utility glass {
  background: var(--glass-bg);
  backdrop-filter: blur(var(--glass-blur))
                   saturate(var(--glass-sat));
  border: 1px solid var(--glass-rim);
  box-shadow: inset 0 1px 0 var(--glass-hi),
              var(--shadow-e3);
}
```

```ts
// motion.ts (Framer Motion / Motion)
export const spring = {
  snappy: { type: "spring", stiffness: 420, damping: 34 },
  gentle: { type: "spring", stiffness: 220, damping: 26 },
  bouncy: { type: "spring", stiffness: 300, damping: 18 },
} as const;
```

ID	REQUIREMENT	PRIORITY
TK-01	Tokens exist in three tiers: **primitive** (espresso-900) → **semantic** (color-text) → **component** (button-primary-bg). Components consume semantic/component tiers only.	MUST
TK-02	Figma Variables use the exact same names and modes (Light / Dark / Night Decaf / High Contrast). A token diff is part of every design PR.	MUST
TK-03	Theme switch must not cause layout shift or a flash of the wrong theme: inline the theme script in `<head>` before paint.	MUST

Motion that pours

Motion should feel like liquid: quick to respond, soft to settle. Interactive elements use springs (interruptible); ambient and scripted sequences use curves.



Motion durations

TOKEN	VALUE	USE
`dur-instant`	100 ms	Press feedback, toggles
`dur-fast`	160 ms	Hover, chip select, tooltips
`dur-base`	240 ms	Toasts, fades, tab switch
`dur-medium`	320 ms	Nav shrink, tab bar minimise
`dur-slow`	480 ms	Sheets, cup fill, page transitions
`dur-slower`	700 ms	Theme cross-fade, hero reveal
`dur-ambient`	1.2–6 s loop	Steam, shimmer, latte-art loader

Easings & springs

TOKEN	VALUE
`ease-standard`	cubic-bezier(.2,0,0,1)
`ease-out-quart`	cubic-bezier(.25,1,.5,1)
`ease-out-expo`	cubic-bezier(.16,1,.3,1)
`ease-in-out-pour`	cubic-bezier(.65,0,.35,1)
`ease-accelerate`	cubic-bezier(.3,0,1,1)
`spring-snappy`	stiffness 420 · damping 34 · mass 1
`spring-gentle`	stiffness 220 · damping 26
`spring-bouncy`	stiffness 300 · damping 18 (stamps, badges)

Choreography rules

ID	REQUIREMENT	PRIORITY
TK-M01	Enter = decelerate (`out-quart / out-expo`); exit = accelerate and 25 % shorter than enter.	MUST
TK-M02	Anything the user can drag, flick or interrupt (sheets, tab bar, cup slider, 3D cup) uses a spring, never a fixed-duration tween.	MUST
TK-M03	Stagger lists at 30–40 ms per item, capped at 8 items (max 320 ms total) so menus never feel slow.	SHOULD
TK-M04	Only animate `transform` , `opacity` and `filter` / `backdrop-filter` radius on small elements. No animated `width/height/top` outside FLIP/layout animations.	MUST
TK-M05	Shared-element transitions (menu card → customizer, cart icon ↔ cart sheet) via Framer Motion `layoutId` or the View Transitions API.	SHOULD
TK-M06	Every motion token has a reduced-motion counterpart (see §9): duration ≤ 150 ms, opacity only.	MUST

Building Liquid Glass in CSS

Apple's Liquid Glass is a real-time optical material: it blurs and bends what is behind it, reflects light at its edges, and adapts its tint to content. On the web we approximate it with six stacked layers.




```css
/* GlassSurface.module.css */
.glass {
  position: relative; isolation: isolate;
  background: var(--glass-bg);
  -webkit-backdrop-filter: blur(20px) saturate(180%);
  backdrop-filter: blur(20px) saturate(180%);
  border: 1px solid var(--glass-rim);
  border-radius: 28px;
  box-shadow:
    inset 0 1px 0 var(--glass-hi),     /* top rim */
    inset 0 -1px 0 rgb(0 0 0/.06),  /* base */
    var(--shadow-e3);
}
/* specular sheen, follows --mx/--my */
.glass::before {
  content: ""; position: absolute;
  inset: 0; border-radius: inherit;
  background: radial-gradient(
    120px circle at var(--mx,30%) var(--my,0%),
    rgb(255 255 255/.45), transparent 60%);
  mix-blend-mode: soft-light;
  pointer-events: none;
}
/* optional refraction (Chromium only) */
@supports (backdrop-filter: url(#x)) {
  .glass--refract {
    backdrop-filter: url(#lg-displace)
      blur(14px) saturate(180%);
  }
}
```

Implementation requirements

ID	REQUIREMENT	PRIORITY
LG-01	One reusable `<GlassSurface variant="clear|frosted|tinted" tint="caramel|matcha|berry|nitro">` primitive; no ad-hoc glass CSS in features.	MUST
LG-02	Glass is used only for floating/chrome layers: nav bar, tab bar, sheets, popovers, toasts, segmented controls, hero CTA card, price pill over photos.	MUST
LG-03	Always ship `-webkit-backdrop-filter` (Safari) + standard property; set `isolation: isolate` to keep blend modes local.	MUST
LG-04	Adaptive tint: sample the average luminance under the bar (precomputed per hero image or via a 1×1 canvas read) and switch bar text between Espresso/Steam with a 240 ms cross-fade.	SHOULD
LG-05	Refraction / lensing via SVG `feDisplacementMap` limited to the hero CTA and tab-bar selection pill, Chromium only, behind `@supports`.	COULD
LG-06	Pointer-reactive specular highlight (CSS vars `--mx/--my`, rAF-throttled) on cards & buttons, desktop only (`hover: hover`).	SHOULD

Glass that never breaks

Glass is expensive: each `backdrop-filter` forces the browser to re-render and blur everything behind it on every frame that changes. Budget it like images.

Fallback ladder

TIER	DETECTED BY	RENDERING
Full Liquid Glass	Modern Chromium, Safari 18+, Firefox 103+	blur + saturate + rim + specular (+ refraction on Chromium)
Reduced transparency	`prefers-reduced-transparency: reduce` or in-app toggle	Opaque surface (`--color-surface` at 96%), keep rim + shadow
No backdrop-filter	`@supports not (backdrop-filter: blur(1px))`	Solid tint at 0.88 opacity + subtle noise texture
Low-power device	`navigator.hardwareConcurrency ≤ 4` or `deviceMemory ≤ 4` or Save-Data	blur capped at 12px, no specular tracking, no refraction
High contrast	`prefers-contrast: more` / `forced-colors: active`	Opaque surfaces, 2px solid borders, system colours (`Canvas` , `CanvasText`)
Print / no-JS	`@media print` , JS disabled	Opaque, no motion; menu and store info fully readable

Performance rules

ID	REQUIREMENT	PRIORITY
LG-P01	Max **3 large** (> 25 % viewport) backdrop-filter surfaces visible at once; small chips don't count but stay ≤ 12px blur.	MUST
LG-P02	Never animate blur radius on large surfaces; animate opacity of a pre-blurred layer instead.	MUST
LG-P03	Promote fixed glass bars with `will-change: transform` only while scrolling; remove after 300 ms idle.	SHOULD
LG-P04	No backdrop-filter inside scrolling lists of > 12 items (menu grid): use pre-baked translucent PNG/AVIF or solid tint.	MUST
LG-P05	Scroll-linked effects use CSS `animation-timeline: scroll()` or Motion `useScroll` (passive, rAF), never scroll event handlers that write layout.	MUST
LG-P06	Glass must hold 60 fps on a mid-range Android (e.g. Pixel 6a class) at 4× CPU throttle; verified in Chrome DevTools Performance panel.	MUST

Do / Don't

Do

- Let product photos scroll *under* the floating nav so the glass visibly picks up colour.
- Keep text on glass ≥ 600 weight for labels < 15px and verify contrast on the worst backdrop.
- Use concentric radii: tab bar 999 → selection pill 999, sheet 28 → inner card 16.

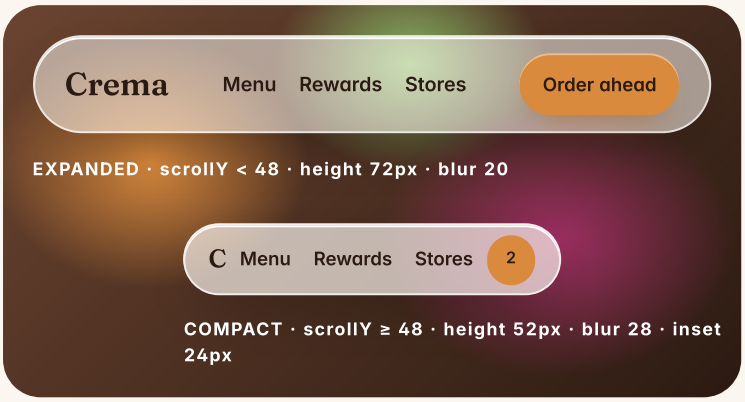
Don't

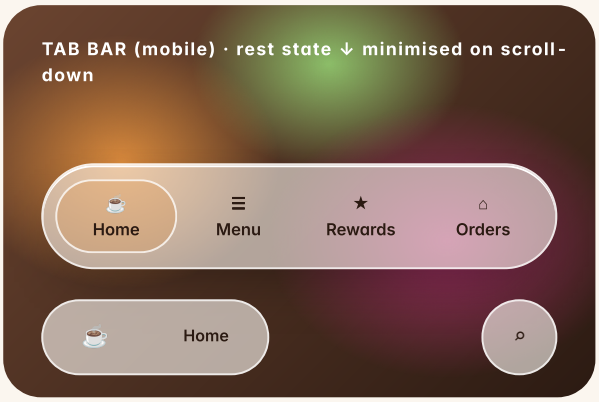
- Stack glass on glass on glass (max 2 glass layers overlapping).
- Put paragraphs of body copy on clear glass.
- Blur the product itself, menu prices, or allergen information.

Browser notes (Oct 2026): `backdrop-filter` is Baseline across evergreen browsers; SVG-filter backdrops (refraction) are Chromium-only, so treat them as enhancement. Test Safari iOS first.

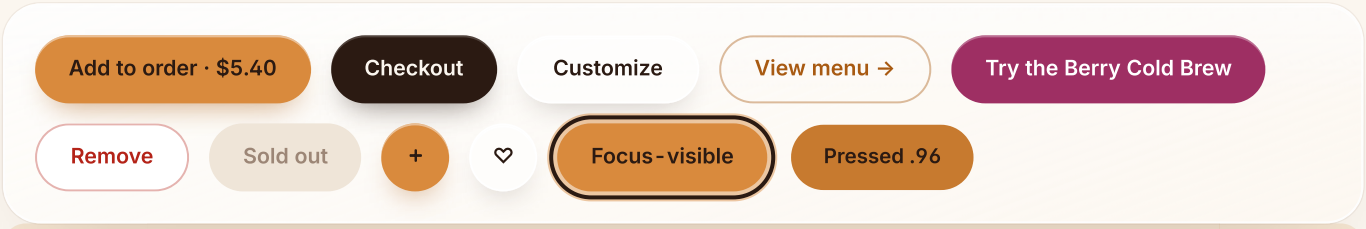
Floating chrome

All chrome floats above content as Liquid Glass and gets out of the way while reading. Components are built on shadcn/ui (Radix primitives) and restyled with our tokens.





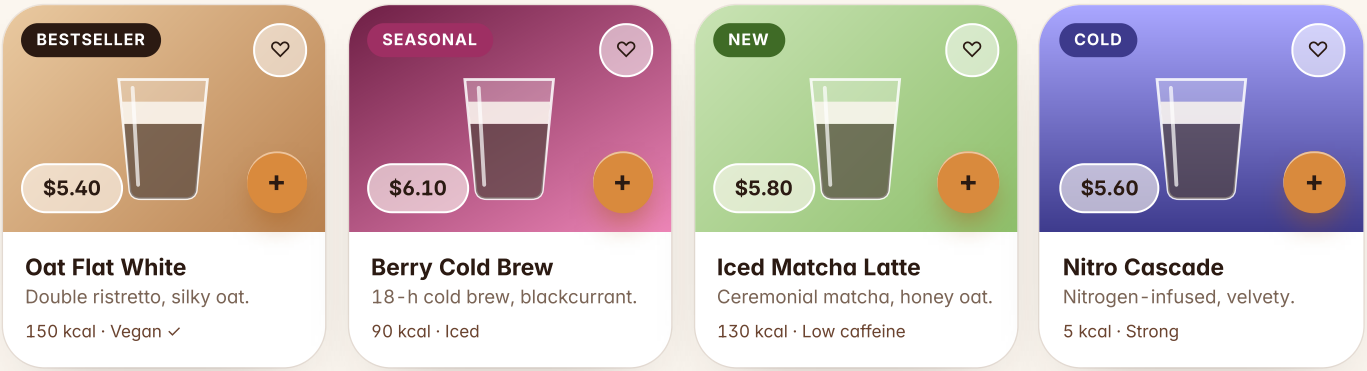
Buttons: variants & states



ID	REQUIREMENT	PRIORITY
CMP-NAV-01	Desktop/tablet top nav: floating glass capsule (radius full, inset 16 px from viewport, height 72 px). On scroll ≥ 48 px it shrinks to 52 px, inset 24 px, logo → monogram, blur 20 → 28 (ANM-02). Restores on scroll-up of ≥ 24 px.	MUST
CMP-NAV-02	Nav contains: logo, Menu, Rewards, Stores, Story, cart button with live count badge, account avatar, primary CTA "Order ahead". Active route uses a sliding glass selection pill (`layoutId`).	MUST
CMP-TAB-01	Mobile (< 768 px): floating bottom tab bar (Home, Menu, Rewards, Orders) 64 px tall, 12 px from bottom + `env(safe-area-inset-bottom)`. On scroll-down it minimises to a compact pill showing only the active tab + separate search button (iOS 26 behaviour); scroll-up or tap restores.	MUST
CMP-TAB-02	An active order shows as a live "accessory" capsule above the tab bar (status + ETA), tapping opens Order Tracking.	SHOULD
CMP-BTN-01	Buttons: primary (Caramel fill), dark, glass, ghost, berry (seasonal), destructive; sizes S 36 / M 44 / L 52 px height; min target 44×44 px; icon buttons circular.	MUST
CMP-BTN-03	Primary CTA on hero and checkout is a **glass-rimmed caramel capsule** with an animated specular sweep every 6 s (paused off-screen & for reduced motion). Only one primary CTA per viewport.	SHOULD
CMP-BTN-04	Buttons emit `navigator.vibrate(8)` on Android where supported for add-to-cart and stamp events; never for navigation.	COULD
CMP-BTN-02	States: hover (+4% brightness, specular sheen), pressed (scale .96, 100 ms), focus-visible dual ring, loading (latte-art spinner replaces label, width locked), disabled (Crema 100 fill, 4.5:1 label).	MUST

Cards you want to taste

Cards are **solid** surfaces (Steamed Milk / Roast Surface) for legibility; glass appears only on the price pill and quick-add button that float over the photo.

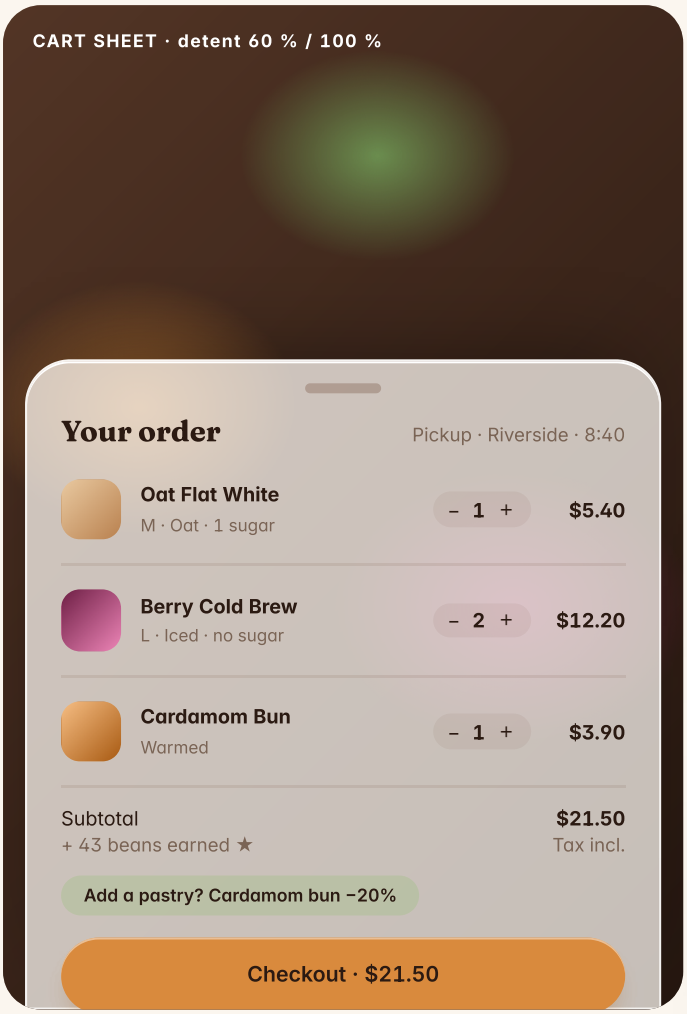


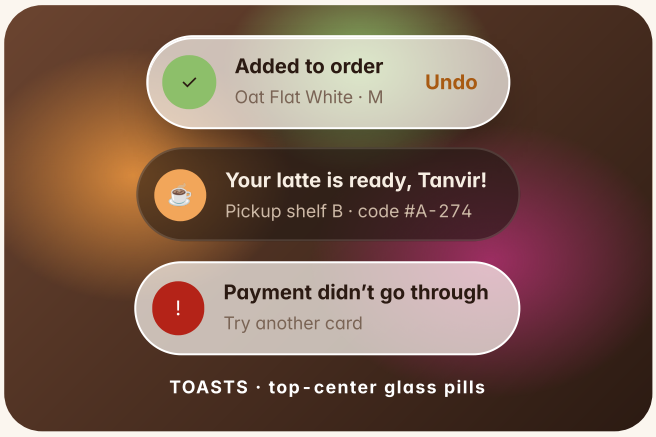


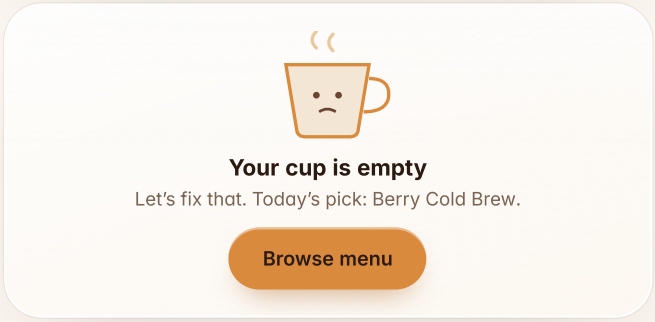
ID	REQUIREMENT	PRIORITY
CMP-CARD-01	Card family: **Menu item** (above), **Promo** (full-bleed gradient + glass CTA), **Info** (store, rewards), **Order** (status). Radius lg 22 px; image ratio 4:3 (grid) or 1:1 (compact list).	MUST
CMP-MENU-01	Menu item card shows: photo/3D render, name, 1-line description, price pill (glass), dietary badges (Vegan, GF, Decaf) with icon + text, quick-add "+" (adds default config) and tap-anywhere → customizer.	MUST
CMP-MENU-02	Sold-out: image desaturated 60 %, "Back at 2 pm" label, quick-add disabled with tooltip. Favourite heart toggles with a 300 ms bounce (MI-04).	SHOULD
CMP-CARD-02	Desktop hover: card lifts 4 px (e2 → e3), photo scales 1.03, specular glare follows pointer; 3D tilt ≤ 4° optional. Whole card is one link; quick-add is a separate button (no nested interactive conflicts).	SHOULD
CMP-CUS-01	Customizer controls: Size segmented control (S/M/L with icon scaling), Milk chips (single), Syrups chips (multi, max 3), Sugar 5-step slider with haptic tick, Shots stepper (0–4), Temperature (Hot/Iced) toggle, Decaf switch, Notes field (140 chars).	MUST
CMP-CUS-03	Allergen & caffeine info (mg) is always visible in the customizer summary and updates with options; never hidden behind glass or animation.	MUST
CMP-CUS-02	Every option change updates price live (tabular numerals, 160 ms roll animation) and the visual cup (SCR-CUST). All controls keyboard operable with arrow keys and labelled for screen readers ("Sugar: Regular, 2 of 4").	MUST

Sheets, toasts & gentle states

Overlays use the thickest glass and the strongest elevation. Loading and empty states are opportunities for personality: no grey boxes, no dead ends.







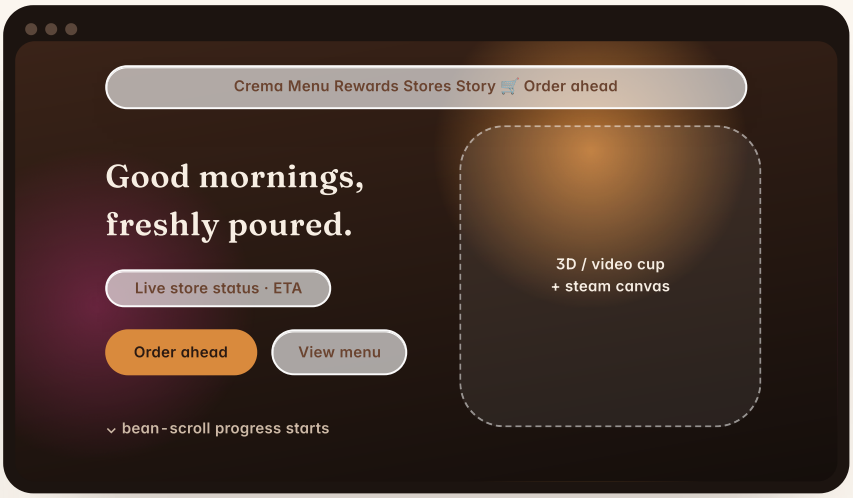
ID	REQUIREMENT	PRIORITY
CMP-CART-01	Cart = bottom sheet on mobile (detents 60 % / 100 %, drag handle, swipe-down to dismiss) and right-side glass drawer (420 px) on ≥ 1024 px. Shows items with thumbnail, options summary, qty stepper, swipe-to-remove + undo toast, subtotal, beans earned, primary Checkout.	MUST
CMP-SHEET-01	Sheets/modals: glass-thick, radius 28 px (top corners on mobile), backdrop scrim 28 % + 16 px blur; focus trapped, Esc / swipe closes, focus returns to trigger (Radix Dialog / vaul Drawer).	MUST
CMP-TOAST-01	Toasts: top-center glass pill, max 2 stacked, auto-dismiss 4 s (errors persist until dismissed), action link (Undo), `role="status"` / `aria-live="polite"`; errors `assertive`. Library: sonner (shadcn).	MUST
CMP-SKEL-01	Skeletons mirror final layout exactly (no CLS), use Crema 100 → Oat 50 shimmer at 1.4 s; replaced with a 160 ms cross-fade. Spinners only for < 1 s actions, using the latte-art loader (FX-03).	MUST
CMP-EMPTY-01	Every empty/error state has: a coffee illustration (cup, bean, steam), a warm one-line message, and a next action. Cases: empty cart, no orders, no rewards yet, store closed, offline, search no results, 404 "This page went decaf".	MUST
CMP-FORM-01	Inputs: 48 px height, radius 16, Crema 100 fill, floating label, inline validation on blur, error text + icon (not colour alone), autocomplete attributes set.	MUST

Home: the first sip

Goal: in 5 seconds a visitor feels the brand, knows if the café is open, and can order. The hero is the single "wow" moment of the site.



DESKTOP ≥ 1280 · HERO SKETCH

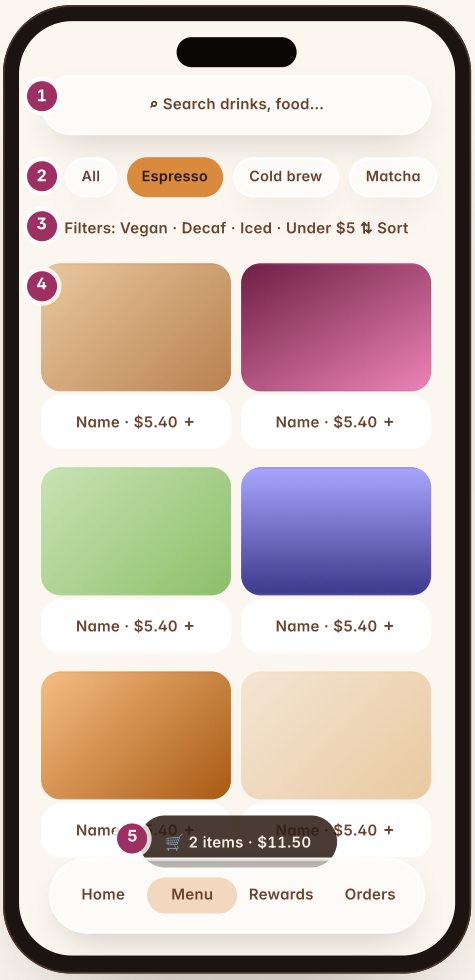


1. Floating glass nav (CMP-NAV-01), transparent over hero, shrinks on scroll.
2. Steam particle hero over cup visual (FX-01); time-of-day greeting.
3. Primary CTA "Order ahead" + secondary "View menu".
4. Live status pill: nearest store, open/closed, current prep ETA.
5. Seasonal specials carousel (snap-scroll, 2.2 cards visible on mobile).
6. Rewards teaser for signed-in users (stamps progress), else join CTA.
7. Floating glass tab bar (CMP-TAB-01).

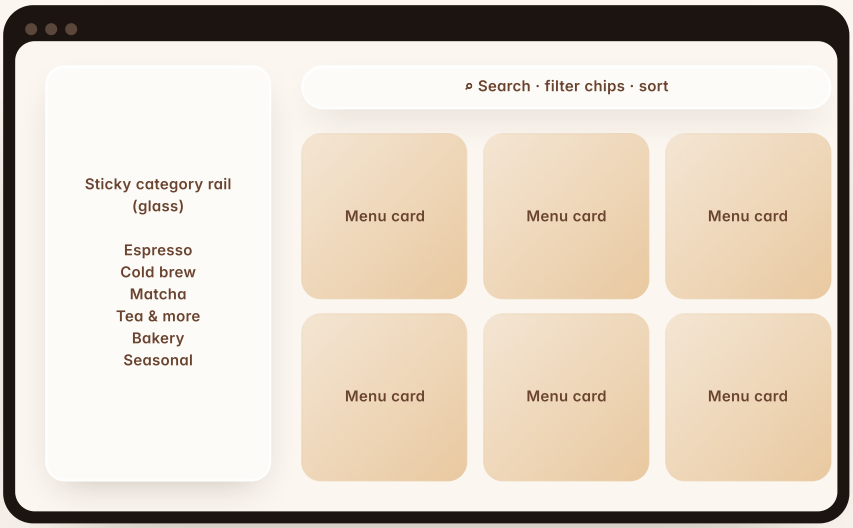
ID	REQUIREMENT	PRIORITY
SCR-HOME-01	Hero fills 100svh on mobile / 88vh desktop with gradient mesh + cup visual; LCP element is the headline text or a ≤ 120 KB AVIF poster, never the canvas.	MUST
SCR-HOME-02	Greeting and hero palette follow time of day (Morning Brew / Afternoon Chill / Night Decaf) using the visitor's local time.	SHOULD
SCR-HOME-03	Below the fold: Seasonal specials → Bestsellers grid → Rewards teaser → Our story strip → Store hours/map preview → Footer (newsletter, socials, legal).	MUST
SCR-HOME-05	Hero must be fully usable with JS disabled or before hydration: headline, CTA and store status are server-rendered HTML.	MUST
SCR-HOME-04	Sections reveal on scroll (fade + 16 px rise, 480 ms, out-quart, once). Bean-scroll progress indicator along the top edge (FX-04).	SHOULD

Menu with categories

Fast to scan, tempting to browse. Categories are sticky glass chips (mobile) or a sticky glass rail (desktop); the grid scrolls underneath so the glass picks up drink colours.



DESKTOP ≥ 1024 · 3-COLUMN GRID + RAIL



1. Sticky glass search; opens full-screen search sheet with recent + popular.
2. Horizontal category chips (scroll-snap); active chip slides with `layoutId`.
3. Dietary filter chips + sort (Popular, Price, Caffeine).
4. Menu cards 2-col mobile / 3-col tablet / 4-col ≥ 1440 (CMP-MENU-01).
5. Floating mini-cart capsule appears after first add; tap → cart sheet.

ID	REQUIREMENT	PRIORITY
SCR-MENU-01	Categories: Espresso, Cold Brew & Nitro, Matcha & Tea, Signature/Seasonal, Bakery, Merch & Beans. Category order changes with time-of-day theme (e.g. Decaf first at night).	MUST
SCR-MENU-02	Scroll-spy: scrolling the list updates the active chip; tapping a chip smooth-scrolls (or jumps, with reduced motion) to the section with sticky section headers.	MUST
SCR-MENU-03	Search is instant (client-side index, < 50 ms), typo-tolerant, matches ingredients (“oat”, “vanilla”) and shows “no results” coffee empty state with suggestions.	SHOULD
SCR-MENU-04	Images lazy-load with blurred LQIP placeholder in brand colours; grid uses CSS `content-visibility: auto` for long lists.	MUST
SCR-MENU-06	Each item exposes structured data (schema.org `MenuItem`, price, dietary) for SEO and rich results.	SHOULD
SCR-MENU-05	Card → customizer uses shared-element transition (image + name morph into the customizer header, 480 ms spring).	SHOULD

The cup that builds itself

The signature screen. A large glass cup on the left (desktop) or top (mobile) **physically fills with layers** as the customer changes options: espresso shots darken the base, milk rises, syrup pools, ice cubes drop, foam settles.



A · Cappuccino S

S · Hot · Whole milk

Microfoam · Steamed milk · Espresso



B · Iced Caramel Oat Latte M

M · Iced · Oat · Caramel

Cold foam · Oat milk · Espresso · Caramel syrup



C · Berry Cold Brew L

L · Iced · Berry cold brew

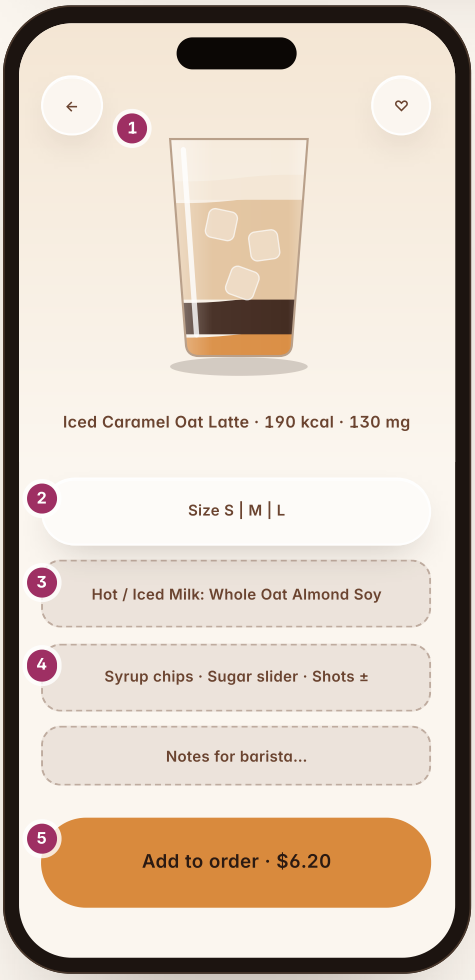
Sweet cream · Cold brew · Berry syrup



D · Iced Matcha M

M · Iced · Matcha

Foam · Matcha · Oat milk



ID	REQUIREMENT	PRIORITY
SCR-CUST-01	Visual cup is an SVG (or R3F canvas when FX-07 is on) with one clip-path; each ingredient is a layer whose height = its share of the recipe volume. Size changes scale the cup (S 0.86 / M 1.0 / L 1.14) with a spring.	MUST
SCR-CUST-02	Layer changes animate: liquid height tween with a sinusoidal surface wave (amplitude 2 px, settles in 500 ms); ice cubes drop with gravity + bounce; foam fades and "settles"; syrup pours as a thin stream before pooling.	MUST
SCR-CUST-03	Hot drinks show steam wisps; iced drinks show condensation droplets (static texture + 2 sliding drops). Temperature toggle cross-fades between them in 320 ms.	SHOULD
SCR-CUST-04	Sticky bottom bar: live price + "Add to order". Price changes roll digit-by-digit (tabular nums). Nutrition & caffeine summary updates live (CMP-CUS-03).	MUST
SCR-CUST-05	Option groups are a single scrolling form on mobile (cup shrinks to 40 % height and docks when scrolling); two-column on desktop with cup sticky left.	MUST
SCR-CUST-06	Presets: "Barista's pick", "Last time", "Less sweet" chips above options; 'Save as favourite' stores the config to the account.	COULD
SCR-CUST-07	Accessibility: the cup is `aria-hidden`; an `aria-live` text summary ("Medium iced oat latte, caramel, 2 shots") is the source of truth.	MUST

Wireframe keys: ① visual cup · ② size segmented control · ③ temperature + milk chips · ④ syrups, sugar, shots · ⑤ sticky add-to-order with live price.

Cart & checkout in 30 seconds

Target: a returning customer re-orders in ≤ 3 taps; a new guest checks out in ≤ 60 s. No forced account creation.

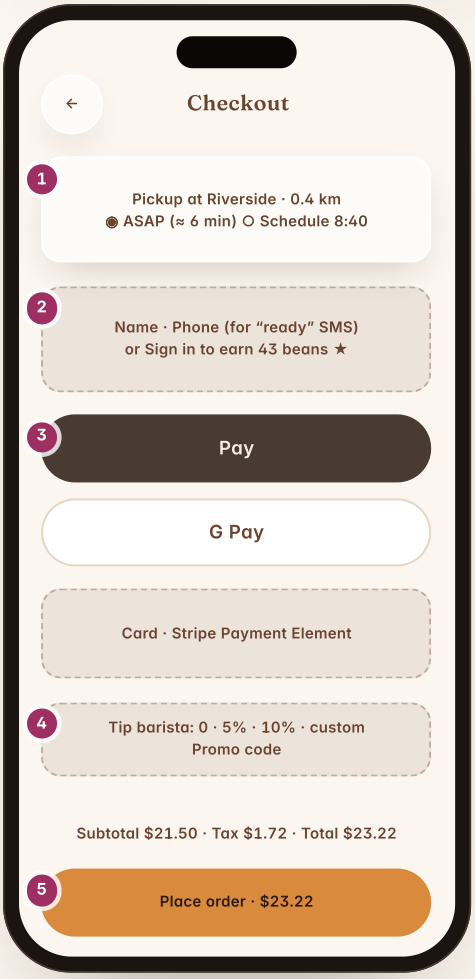
STEP 1
Cart
Review, edit, upsell 1 pastry

STEP 2
Pickup
Store, ASAP or time slot

STEP 3
Details
Guest name + phone/email or sign in

STEP 4
Pay
Apple Pay, Google Pay, card, gift card

STEP 5
Confirmed
Code, ETA, add to wallet



ID	REQUIREMENT	PRIORITY
SCR-CART-01	Cart sheet as CMP-CART-01; editing an item reopens the customizer pre-filled; quantity change animates line total; remove = swipe or trash with Undo toast (5 s).	MUST
SCR-CART-02	Smart upsell: max one contextual suggestion ("Pairs well: Cardamom bun") as a compact glass card; dismissible, never blocks checkout.	SHOULD
SCR-CHK-01	Single-page checkout with 4 collapsible glass sections (Pickup, Details, Payment, Review); completed sections collapse to a one-line summary with an edit link.	MUST
SCR-CHK-02	Express wallets (Apple Pay / Google Pay via Payment Request API) shown first when available; card via a PCI-compliant hosted element (e.g. Stripe). No card data touches our servers.	MUST
SCR-CHK-03	Pickup scheduler: ASAP with live ETA, or 10-minute slots for today/tomorrow within store hours; full slots disabled with reason.	MUST
SCR-CHK-04	WCAG 2.2: no redundant entry (3.3.7, remember details), accessible authentication (3.3.8, passkeys / magic link, no puzzles), all errors announced and linked to fields.	MUST
SCR-CHK-05	"Place order" triggers a pour animation filling the button (600 ms) then navigates to tracking; on failure the liquid drains and an error toast appears.	SHOULD
SCR-CHK-06	Confirmation shows pickup code (JetBrains Mono, large), QR, ETA, "Add to Apple/Google Wallet", and receipt email.	SHOULD

Wireframe keys: ① pickup store & time · ② guest details or sign-in · ③ express wallets · ④ tip & promo · ⑤ sticky place-order.

Watch it brew

Real-time status turns waiting into anticipation. A progress ring wraps a cup that fills stage by stage while steam intensifies; the final "Ready" moment celebrates.



STAGE 3 / 5 · BREWING

- ☑ **Received** · order accepted by store, cup appears empty — 08:31
- ⚙ **Grinding** · beans icon spins, grinder hum (if sound on) — 08:32
- ☕ **Brewing** · espresso layer pours in, ring 60 % — 08:34
- ♨ **Steaming** · milk layer rises, steam doubles — –
- ★ **Ready** · cup glows, confetti of coffee beans, haptic + push — –

> **Data:** server-sent events or WebSocket from the POS/KDS; fall back to 15 s polling. ETA recalculated per stage; never shows a stage the barista hasn't reached.

ID	REQUIREMENT	PRIORITY
SCR-TRK-01	Five stages (Received, Grinding, Brewing, Steaming/Finishing, Ready) with timestamps; ring progress animates between stages over 800 ms (ease-in-out-pour); idle shimmer on the active arc.	MUST
SCR-TRK-02	Cup illustration mirrors the actual drink (reuses SCR-CUST layer data) and fills to the stage's percentage.	SHOULD
SCR-TRK-03	Ready state: Lottie bean-burst (≤ 1.2 s, ≤ 60 KB), pickup code & shelf number enlarged, optional Web Push + vibration; screen stays awake (Wake Lock API) while on page.	SHOULD
SCR-TRK-04	Status is mirrored in the tab-bar accessory capsule (CMP-TAB-02) and the document title ("☕ Brewing · Crema").	SHOULD
SCR-TRK-05	Accessible: status changes announced via `aria-live="polite"`; reduced motion replaces ring animation with instant step updates; colour is never the only stage indicator (icons + text).	MUST
SCR-TRK-06	Help actions: "I'm running late" (+10 min hold), "Call store", "Get directions"; cancel allowed only before Grinding.	COULD

READY STATE · DARK GLASS

Your latte is ready ☕

PICKUP CODE **#A-274**

SHELF **B**

TAB-BAR LIVE CAPSULE (CMP-TAB-02)

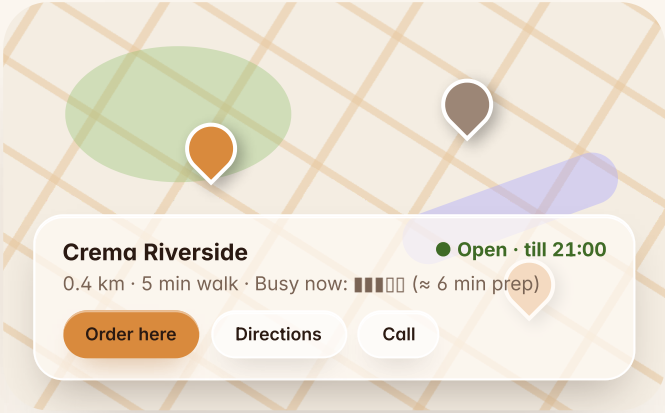
Brewing · Oat Flat White · 3 min

Home · Menu · Rewards · Orders

Rewards card & store locator

Rewards should feel like a beautiful physical object you want to collect; the locator answers "Is it open, how far, how busy?" at a glance.





Loyalty / rewards

ID	REQUIREMENT	PRIORITY
SCR-LOY-01	Card is a 3D-tilting glass/metal object (pointer & gyroscope on mobile, ≤ 8°), tier-coloured: Oat (base), Caramel (silver), Gold Crema, Berry Reserve.	MUST
SCR-LOY-02	Stamp grid (10) + beans balance; new stamp animates as a pressed ink stamp with a spring bounce (ANM-13) and a soft "thunk" sound if ambient audio is on.	MUST
SCR-LOY-03	Rewards catalogue (redeem drinks, pastries, beans); member QR for in-store scan; Add to Apple/Google Wallet.	SHOULD
SCR-LOY-04	Challenges & streaks (e.g. "Try 3 cold brews this month") shown as progress pills.	COULD

Store locator & hours

ID	REQUIREMENT	PRIORITY
SCR-LOC-01	Map (MapLibre GL + custom warm style matching tokens) with glass bottom-card list; list-only view by default when geolocation is denied.	MUST
SCR-LOC-02	Each store: open/closed state with countdown ("Closes in 40 min"), weekly hours table, holiday exceptions, address, amenities (Wi-Fi, outlets, accessible entrance, pet-friendly).	MUST
SCR-LOC-03	Live busyness meter & prep ETA from POS; "Order here" deep-links checkout with store preselected.	SHOULD
SCR-LOC-04	Map lazy-loads only on interaction/visibility; static map image placeholder first (PERF-05).	MUST

TIER LADDER · card materials

Oat	Caramel	Gold Crema	Berry Reserve
0+ ★	300+ ★	1,000+ ★	2,500+ ★

HOURS WIDGET · today highlighted

Mon – Tue	07:00 – 20:00
Wed	**07:00 – 21:00 · open now**
Thu – Fri	07:00 – 22:00 · late decaf bar
Sat	08:00 – 22:00
Sun	08:00 – 18:00

Holiday: 24 Dec 08:00 – 14:00 · uses `schema.org/OpeningHoursSpecification`

Our story & pickup reservations

The About page is editorial and scroll-driven, a little magazine about bean origins and people. Pre-order lets guests reserve a pickup slot or a table for groups.





About / story

1. Hero origin story with slow parallax
2. Scroll-scrubbed farm-to-cup timeline (GSAP ScrollTrigger)
3. Values & sustainability glass cards

Pre-order / reservations

1. Mode switch: Pickup slot vs. table
2. Date strip (7 days)
3. 10-min slots; full slots struck out
4. Optional drink pre-order timed to slot

ID	REQUIREMENT	PRIORITY
SCR-ABT-01	Editorial layout: max 68ch text, Fraunces display, full-bleed imagery; chapters (Origin, Roast, People, Planet) with scroll-linked reveals; reduced motion = static stacked layout.	MUST
SCR-ABT-02	Farm-to-cup timeline scrubbed by scroll (bean travels along a path; FX-04 shares the asset).	SHOULD
SCR-PRE-01	Pickup pre-order: choose store → date (7-day strip) → 10-min slot → optional drinks → confirm; slot capacity from backend, optimistic hold for 5 min.	MUST
SCR-PRE-02	Table reservations for groups (2–12), indoor/terrace, with confirmation email + .ics calendar file + cancel/modify link.	SHOULD
SCR-PRE-03	Reminders: push/email 30 min before slot; “Running late” extends hold +10 min once.	COULD

07 · CREATIVE COFFEE CONCEPTS

Ideas that make it unforgettable

Signature moments that turn a menu website into a sensory experience. Each idea has an owner priority; all Could items ship behind feature flags and respect reduced motion, data saver and low-power tiers.



FX-01 MUST

Steam particle hero

Canvas 2D (or Three.js Points) wisps rising from the hero cup, curl-noise drift, react gently to cursor/scroll. 40–80 particles, pauses off-screen. Static SVG steam fallback.



FX-02 MUST

Pour & fill animations

Liquid fills: customizer cup, “Place order” button, add-to-cart badge. Shared SVG wave component with spring-driven level.



FX-03 SHOULD

Latte-art loader

Rosetta/heart drawn by animated stroke-dashoffset inside a crema circle, 1.2 s loop. Replaces all generic spinners.



FX-04 SHOULD

Bean-scroll progress

A coffee bean travels and rotates along a 2 px caramel line at the top of long pages; roast colour darkens as you progress.



FX-05 SHOULD

Morning brew vs night decaf

Theme + copy + menu order follow local time (3 modes). Manual override in settings; transition via View Transitions cross-fade.



FX-06 COULD

Ambient café sound

Opt-in toggle in nav: espresso machine, soft chatter, rain-on-window mix (Web Audio, 3 loops ≤ 300 KB). Never autoplays; remembers choice; volume slider.



FX-07 COULD

AR-like 3D cup preview

React Three Fiber glTF cup (≤ 500 KB, Draco) with your exact layers; drag to rotate, idle float, glass refraction shader. Optional WebXR “place on table”.



FX-08 SHOULD

Seasonal theme packs

Pumpkin Spice (autumn), Berry Cold Brew (winter), Matcha Bloom (spring), Nitro Summer: swap accent tokens, hero art and specials via CMS flag.



FX-09 COULD

Crema ripple hover

On desktop, hovering a drink photo sends a subtle concentric ripple (CSS radial mask) like tapping a latte; buttons get a caramel specular sweep.



FX-10 COULD

Brew-your-mood finder

3-question playful quiz (energy, taste, temperature) with swipeable glass cards → recommended drink with pre-filled customizer.



FX-11 COULD

Flavour wheel & origin map

Interactive SVG flavour wheel for single-origin beans (fruity, chocolatey, floral); selecting a note highlights origins on a stylised map.



FX-12 COULD

Weather-aware picks

Light-weight weather lookup (by store city) re-orders specials: hot day → iced & nitro first, rainy → spiced lattes, with a tiny weather glyph.

Creative guardrails: one signature effect per screen (BR-08) · every effect has a static equivalent · no effect may delay first input or push LCP · audio is always opt-in · effects pause when the tab is hidden (`visibilitychange`) or element is off-screen (`IntersectionObserver`).

Animation spec table

Normative values for implementation. Durations are for the enter phase unless two values are given (enter / exit). The last column defines the mandatory `prefers-reduced-motion: reduce` behaviour.

ID	TRIGGER	EFFECT	DURATION	EASING	LIBRARY	REDUCED MOTION
ANM-01	Hero in view	Steam particles rise & drift (FX-01)	`loop 4-6 s`	linear + curl noise	Canvas 2D / Three.js	Static SVG steam
ANM-02	Scroll ≥ 48 px	Nav bar shrinks 72→52 px, blur 20→28, logo → monogram	`320 ms`	spring-snappy	Framer Motion useScroll	Instant swap, no scale
ANM-03	Scroll down / up	Tab bar minimises to pill / restores	`320 ms`	spring-snappy	Framer Motion	Opacity only, 150 ms
ANM-04	Press (any button)	Scale .96 + highlight shift; release springs back	`100 / 220 ms`	out-quart	FM whileTap	Colour change only
ANM-05	Add to cart	Thumbnail flies along bezier to cart; badge bumps 1→1.2→1	`600 ms`	cubic-bezier(.2,.8,.2,1)	FM layoutId	Badge count fade
ANM-06	Customizer option change	Cup layers re-level with surface wave; ice drops	`500 ms`	spring-gentle	FM + SVG	Instant level, no wave
ANM-07	Milk / syrup selected	Pour stream then swirl into layer	`900 ms`	in-out-pour	Lottie (dotLottie)	Skip, instant layer
ANM-08	Loading > 300 ms	Latte-art rosetta stroke draw (FX-03)	`1.2 s loop`	in-out-pour	SVG / Lottie	Static icon + “Loading…”
ANM-09	Page scroll	Bean travels & rotates on top progress line (FX-04)	`scroll-linked`	linear	CSS scroll-timeline / FM	Plain progress bar
ANM-10	Order stage change	Ring arc fills, cup level rises, steam intensifies	`800 ms`	in-out-pour	GSAP timeline / FM	Step text updates
ANM-11	Order ready	Bean-burst confetti, cup glow, code scales in	`1.2 s`	out-expo	Lottie	Static “Ready” badge
ANM-12	Sheet open / close	Slide up from detent; scrim blur 0→16 px	`380 / 280 ms`	spring 300/30	vaul + FM	Fade 150 ms

Animation spec, continued

ID	TRIGGER	EFFECT	DURATION	EASING	LIBRARY	REDUCED MOTION
ANM-13	Toast	Glass pill drops from top, auto-hide	240 / 180 ms	out-expo	sonner	Fade
ANM-14	Stamp earned	Stamp presses with ink splat; card tilts	450 ms	spring-bouncy	FM	Stamp appears, no tilt
ANM-15	Category switch	Active pill slides; grid cross-fades with 30 ms stagger	300 ms	spring-snappy	FM layoutId	No stagger, fade
ANM-16	Theme / time-of-day switch	Token cross-fade via View Transitions	600 ms	standard	View Transitions API	Instant
ANM-17	Content loading	Skeleton crema shimmer sweep	1.4 s loop	linear	CSS keyframes	Static skeleton
ANM-18	Card hover (desktop)	Lift 4 px, shadow e2→e3, glare follows pointer	200 ms	out-quart	CSS + FM	Shadow only
ANM-19	Section enters viewport	Fade + 16 px rise, once	480 ms	out-quart	FM whileInView	Fade 150 ms / none
ANM-20	Menu → customizer	Shared element morph of image & title	480 ms	spring-gentle	FM layoutId / View Transitions	Cross-fade
ANM-21	3D cup idle / drag	Float ±4 px, rotate with inertia	loop 3 s	spring 120/14	Three.js + R3F + drei	Static render (AVIF)

Library responsibilities

Framer Motion (Motion)
Default for all UI motion: springs, gestures, `layoutId`, `useScroll`, `AnimatePresence`, `MotionConfig reducedMotion="user"`.

GSAP + ScrollTrigger
Optional: long scroll-scrubbed storytelling (About timeline) and the multi-step brew timeline. Lazy-loaded per route.

Lottie (dotLottie)
Designer-authored illustrations: pour, bean-burst, latte-art. Each ≤ 60 KB, ≤ 60 fps, played via IntersectionObserver.

Three.js / R3F
Optional 3D cup (FX-07) and WebGL steam. Dynamic import on idle + interaction; never in the critical path; ≤ 250 KB gz.

```tsx
// Global motion config (app/providers.tsx)
import { MotionConfig } from "motion/react";
export function Providers({ children }) {
  return (
    <MotionConfig reducedMotion="user"
      transition={spring.snappy}>
      {children}
    </MotionConfig>
  );
}
```

```css
/* CSS-level safety net */
@media (prefers-reduced-motion: reduce) {
  *, *::before, *::after {
    animation-duration: 0.01ms !important;
    animation-iteration-count: 1 !important;
    transition-duration: 150ms !important;
    scroll-behavior: auto !important;
  }
  .fx-ambient { display: none; }
}
```

Small moments, big feelings

Micro-interactions confirm every action within 100 ms and add warmth. Each is tiny, interruptible and has a reduced-motion equivalent.

ID	REQUIREMENT	PRIORITY	
MI-01	Quantity stepper	Digits roll vertically; at max qty the stepper does a 2-px “no” shake	MUST
MI-02	Price change	Tabular digits roll; total briefly tints Caramel 700 (160 ms)	MUST
MI-03	Add to cart	Button fills with coffee (pour), label morphs “Add” → “Added ✓”, cart badge bump	MUST
MI-04	Favourite	Heart pops with 6 tiny bean particles; haptic tick on mobile	SHOULD
MI-05	Size select	Cup icon grows/shrinks; selection pill slides (layoutId)	MUST
MI-06	Sugar slider	Sugar cubes drop into cup per step; detents snap with haptic	SHOULD
MI-07	Hot ↔ Iced toggle	Steam evaporates / ice cubes clink in; thumb leaves frost trail	SHOULD
MI-08	Pull to refresh (orders)	Pull reveals a filling espresso cup; release = latte-art loader	COULD
MI-09	Form validation	Valid field gets a small matcha check; error gently shakes once + message	MUST
MI-10	Copy pickup code	Code flashes, “Copied” glass tooltip, 1.5 s	SHOULD
MI-11	Theme toggle	Sun ↔ moon icon morphs as a coffee cup ↔ decaf moon	COULD
MI-12	Sound toggle	Speaker icon with steam waves that animate only when sound is on	COULD

Reduced-motion & comfort rules

ID	REQUIREMENT	PRIORITY
RM-01	Honour `prefers-reduced-motion: reduce` globally (MotionConfig + CSS safety net) and offer an in-app “Reduce motion” toggle stored in localStorage/account.	MUST
RM-02	With reduced motion: no parallax, no scroll-scrubbed animation, no particle/3D effects, no auto-playing carousels; transitions become ≤ 150 ms opacity fades.	MUST
RM-03	Never flash more than 3 times per second (WCAG 2.3.1); no full-screen flashes on “Ready”.	MUST
RM-04	Any animation > 5 s (ambient steam, shimmer, sound visualiser) can be paused, stopped or hidden (WCAG 2.2.2); ambient loops stop after 30 s of inactivity.	MUST
RM-05	Motion triggered by interaction (parallax, tilt) can be disabled (WCAG 2.3.3, AAA, adopted as house rule).	SHOULD
RM-06	Low-power / Save-Data tier disables FX-01, FX-06, FX-07 and caps blur (LG-P fallback ladder).	SHOULD

Mobile-first, everyone-first

Design starts at 390 × 844 and scales up. Accessibility is a launch blocker, not a polish task: target **WCAG 2.2 AA** across all flows.

xs	0–379 px	4 col · 16 gutter	Small phones; tab bar labels hidden
sm	380–767 px	4 col · 16	Default phone layout (design base 390)
md	768–1023 px	8 col · 24	Tablet; nav bar replaces tab bar
lg	1024–1279 px	12 col · 24	Cart as side drawer; 3-col menu
xl	1280–1535 px	12 col · 32	Desktop base (1440 frame)
2xl	≥ 1536 px	12 col · 32, max 1440	Content max-width, ambient bg extends

Responsive requirements

ID	REQUIREMENT	PRIORITY
RSP-01	Mobile-first CSS (min-width queries only); container queries for cards & customizer panels.	MUST
RSP-02	Test widths 360, 390, 430, 768, 1024, 1280, 1440, 1920; landscape phone keeps checkout CTA visible.	MUST
RSP-03	Use `svh/dvh` units for hero & sheets; respect safe-area insets (notch, home indicator).	MUST
RSP-04	Touch targets ≥ 44 × 44 px on touch; hover effects only under `(hover:hover) and (pointer:fine)`.	MUST
RSP-05	Images via `next/image` with `sizes`, AVIF/WebP, art-directed crops per breakpoint.	MUST

Accessibility requirements

ID	REQUIREMENT	PRIORITY
A11Y-01	Contrast per §2 matrix; text on glass validated on worst-case backdrop (TK-C05).	MUST
A11Y-02	Full keyboard support; visible dual-ring focus; focus never hidden under sticky glass bars (SC 2.4.11) via `scroll-padding`.	MUST
A11Y-03	Semantic landmarks, one `h1` per page, skip link, Radix ARIA patterns for dialogs, tabs, sliders, menus.	MUST
A11Y-04	Target size ≥ 24 px minimum (SC 2.5.8), 44 px house rule; dragging actions (sheets, slider) have click alternatives (SC 2.5.7).	MUST
A11Y-05	Screen-reader text for visual-only content (cup, ring, stamps); live regions for cart, price, order status.	MUST
A11Y-06	Supports 200 % zoom & 400 % reflow, `forced-colors`, `prefers-contrast`, reduced transparency.	MUST
A11Y-07	Allergen/dietary info as text + icon; never colour-only. Language attr set; i18n-ready strings.	MUST

Fast as an espresso shot

Beauty must never cost speed. Budgets are measured at p75 on real users (CrUX/RUM) and in CI with Lighthouse on a throttled mid-range mobile profile.

LCP	INP	CLS	FPS	JS	Lighthouse
< 2.5 s	< 200 ms	< 0.1	60	≤ 170 KB	≥ 90
target 1.8 s	target 120 ms	target 0.02	all animations	gz initial route	perf · a11y 100

ID	REQUIREMENT	PRIORITY
PERF-01	Core Web Vitals budgets above are release gates; CI fails if Lighthouse CI or bundle-size check regresses > 5 %.	MUST
PERF-02	Hero: server-rendered text + AVIF poster ≤ 120 KB with `fetchpriority="high"`; steam canvas starts after `requestIdleCallback`.	MUST
PERF-03	Heavy effects (Three.js, GSAP, Lottie, Web Audio, map) are dynamically imported, only on visibility/interaction, and skipped on Save-Data / low-power tier.	MUST
PERF-04	Fonts: ≤ 3 WOFF2 files, ≤ 100 KB total, preloaded, subset; images lazy below the fold; menu JSON ≤ 60 KB gz.	MUST
PERF-05	Animate only compositor-friendly properties; ≤ 3 large backdrop-filter layers; no long tasks > 50 ms during scroll.	MUST
PERF-06	Offline & flaky network: PWA service worker caches shell, menu and last order; graceful offline empty state.	SHOULD

Suggested tech stack

FRAMEWORK
Next.js (latest stable, App Router, RSC, Server Actions)

LANGUAGE
TypeScript (strict), ESLint, Prettier

STYLING
Tailwind CSS v4 + CSS variables tokens, `@utility glass`

COMPONENTS
shadcn/ui on Radix primitives, vaul (drawer), sonner (toasts), cmdk (search)

MOTION
Framer Motion (motion/react); optional GSAP + ScrollTrigger, dotLottie, Three.js + R3F + drei

STATE & DATA
TanStack Query, Zustand (cart), Zod schemas, SSE/WebSocket for order status

MAPS & PAY
MapLibre GL, Stripe Payment Element + Payment Request (Apple/Google Pay)

QUALITY
Storybook, Vitest, Playwright + axe-core, Chromatic visual tests, Lighthouse CI

OPS
Vercel (or similar) edge, image CDN, RUM via web-vitals, feature flags for FX-*

12 · ACCEPTANCE CRITERIA

How we know it’s done

Given / When / Then criteria for the critical journeys. A screen is “done” only when its Must requirements pass, the QA checklist is green and the owner signs off.

ID	JOURNEY	CRITERION (GIVEN / WHEN / THEN)	TRACES TO
AC-01	**Home hero**	Given a first-time mobile visitor on 4G, when the home page loads, then the headline + CTA render < 2.5 s (LCP), store status is visible without scrolling and the steam effect starts only after first paint.	SCR-HOME-01/05, PERF-02
AC-02	**Glass nav**	Given any page, when the user scrolls ≥ 48 px, then the nav shrinks to compact within 320 ms without layout shift, text keeps ≥ 4.5:1 on all hero images, and scrolling up restores it.	CMP-NAV-01, TK-C05
AC-03	**Customizer**	Given the customizer, when any option changes, then the cup layers, price and nutrition update within 100 ms of input and the animation completes ≤ 500 ms; screen readers announce the new summary.	SCR-CUST-01/02/07
AC-04	**Add to cart**	Given a drink, when “Add to order” is pressed, then visual feedback starts < 100 ms, cart count increments, an Undo toast appears and focus stays on the triggering button.	MI-03, CMP-TOAST-01
AC-05	**Checkout**	Given a guest with Apple/Google Pay, when they checkout, then an order can be placed in ≤ 4 interactions and ≤ 60 s, with no account required and no card data handled by our servers.	SCR-CHK-01/02
AC-06	**Tracking**	Given a placed order, when the store updates the status, then the UI reflects the new stage within 3 s (SSE) or 15 s (poll fallback) and announces it via aria-live.	SCR-TRK-01/05
AC-07	**Reduced motion**	Given reduce-motion is on, when browsing any flow, then no parallax, particles, 3D or scroll-scrub runs, and all transitions are ≤ 150 ms fades.	RM-01/02
AC-08	**Themes**	Given local time 19:30 and no manual choice, when the site loads, then Night Decaf theme is applied before first paint (no flash) and decaf items appear first.	FX-05, TK-03
AC-09	**Accessibility**	Given automated axe scans and manual keyboard + VoiceOver/TalkBack passes, then there are 0 serious/critical violations on all key screens.	A11Y-01…07
AC-10	**Performance**	Given Lighthouse CI (mobile, Moto G Power profile), then Performance ≥ 90, Accessibility = 100, Best Practices ≥ 95 on Home, Menu, Customizer, Checkout.	PERF-01

Design QA checklist

Run per screen at 390, 768 and 1440 px, in light, dark and Night Decaf, before every design hand-off and again on the staging build (pixel-diff with Chromatic / Figma overlay).

Visual & tokens

- [ ] Only token colours, fonts, radii and shadows used (no raw hex)
- [ ] Concentric radii respected in nested surfaces
- [ ] Type scale and line lengths match spec
- [ ] Spacing on the 4-pt grid; gutters per breakpoint
- [ ] Icons from one set (e.g. Phosphor / Lucide), 1.5 px stroke, optical sizes

Liquid Glass

- [ ] Glass only on chrome / overlays (LG-02)
- [ ] ≤ 3 large blur surfaces visible (LG-P01)
- [ ] Rim + highlight + shadow present on all glass
- [ ] Legible on lightest and darkest backdrop
- [ ] Fallbacks verified: reduced transparency, no-support, high contrast

Motion

- [ ] Durations / easings match ANM table
- [ ] Interruptible gestures use springs
- [ ] Reduced-motion variant reviewed for every animation
- [ ] 60 fps in DevTools at 4× CPU throttle
- [ ] Ambient effects pause off-screen / hidden tab

Accessibility

- [ ] Contrast checked (text 4.5:1, UI 3:1)
- [ ] Focus order logical; focus visible & unobscured
- [ ] Targets ≥ 44 px; drag has click alternative
- [ ] Screen-reader labels for cup, ring, stamps, prices
- [ ] 200 % zoom / 400 % reflow without loss

Content & states

- [ ] Loading, empty, error, offline, sold-out states designed
- [ ] Long names, 3-line descriptions and 5-digit prices tested
- [ ] Microcopy warm, concise, consistent
- [ ] Allergens and caffeine visible as text
- [ ] Localisation room (+30 % text length)

Responsive & performance

- [ ] Layouts at all 6 breakpoints + landscape
- [ ] Safe-area insets on iOS / Android
- [ ] Images sized, AVIF, lazy below fold
- [ ] Heavy FX lazy-loaded and flagged
- [ ] CWV budgets pass in Lighthouse CI

Severity for QA findings: S1 blocks release (Must failing, a11y blocker, payment issue) · **S2** fix before launch (visual defect on key screen, motion jank) · **S3** polish backlog. Log findings with requirement ID, screenshot, breakpoint, theme and browser.

What design delivers

Everything lives in one Figma project with a shared library. Developers should never have to guess a value: if it is not in Figma Variables or this document, ask.

ID	DELIVERABLE	PRIORITY
DLV-01	**Foundations file.** Figma Variables for colour (Light / Dark / Night Decaf / High Contrast modes), type, spacing, radius, elevation, blur, motion; exported via Tokens Studio → Style Dictionary JSON.	MUST
DLV-02	**Component library.** All CMP-* components with variants, states (default, hover, pressed, focus, disabled, loading, error), auto-layout, responsive props, dark mode; usage notes per component.	MUST
DLV-03	**Liquid Glass kit.** Glass surface styles (clear, frosted, tinted ×4), rim/highlight effects, background meshes, plus annotated "do / don't" page.	MUST
DLV-04	**Key screens.** Hi-fi frames for all SCR-* screens at 390, 768, 1440 px; light + dark; plus empty/loading/error states.	MUST
DLV-05	**Prototypes.** Clickable flows: browse → customize → cart → checkout → tracking → ready; rewards; pre-order. Smart-animate approximations of signature motion.	MUST
DLV-06	**Motion specs.** After Effects / Rive / Lottie files for ANM-01, 06, 07, 08, 10, 11, 14 with exported dotLottie (≤ 60 KB each) and a timing sheet matching §8.	SHOULD
DLV-07	**Illustration & 3D assets.** Cup layer SVG kit, steam, beans, empty-state illustrations (SVG), optional glTF cup (≤ 500 KB, Draco) with materials.	SHOULD
DLV-08	**Photography direction.** Shot list & style: warm natural light, glass vessels, layered drinks on oat/stone backgrounds; 4:3 & 1:1 crops; AVIF masters.	SHOULD
DLV-09	**Accessibility annotations.** Focus order, landmarks, ARIA roles, alt text, live regions, reduced-motion variants annotated on every screen.	MUST
DLV-10	**Dev Mode hand-off.** Frames marked "Ready for dev", Code Connect mappings to shadcn components, changelog per release.	SHOULD

Suggested design timeline

W1
Discovery & mood
Moodboard, tokens draft

W2–3
Foundations & glass kit
DLV-01, DLV-03

W3–5
Components
DLV-02, Storybook parity

W5–7
Screens & prototypes
DLV-04, DLV-05

W7–8
Motion & assets
DLV-06, DLV-07

W9
QA & sign-off
Checklist green, AC pass

DOCUMENT OWNER	DESIGN LEAD	FRONTEND LEAD
Tanvir Hasan		
Signature · date	Signature · date	Signature · date

Changes to Must requirements need owner approval and a version bump (v1.1, v1.2…). Brand name "Crema" is a placeholder. Liquid Glass is an Apple design language; this project draws inspiration from it and doesn't copy Apple assets or SF Pro fonts.